

■ SignLens Backend Gateway Subsystem

Node.js Express API Gateway, Database Setup & XAMPP Integration Guide

Welcome to the **SignLens Backend Gateway Subsystem**. This server acts as the central API Gateway and security layer for the entire SignLens platform. It manages user authentication (JWT & bcrypt), database transactions via Knex.js ORM, translation history logging, Cloudinary media management, Learning Management System (LMS) courses, and proxies prediction payloads to the Python ML server.

■ 1. Directory Structure Overview

```
SignLens/backend/
■■■ src/
■   ■■■ config/           # Environment & Knex database configuration
■   ■■■ controller/       # Express route handlers (auth, predict, history, lesson, quiz, user)
■   ■■■ database/         # Database migration & seed files
■   ■   ■■■ migrations/   # Knex SQL table creation schemas
■   ■   ■■■ seed/         # Seed data for lessons, quizzes, badges
■   ■■■ docs/             # Swagger OpenAPI documentation JSON specifications
■   ■■■ middleware/       # Auth verification middleware (JWT validation)
■   ■■■ model/            # Database models (Knex/SQL wrappers)
■   ■■■ routes/           # Express API route declarations
■   ■■■ services/         # ML Proxy client (mlClient.service.js) & WebSocket stream service
■   ■■■ main.js           # Express application entry point
■■■ .env.example          # Environment variable template
■■■ knexfile.js           # Knex.js database connection config
■■■ package.json          # Node.js dependencies & execution scripts
```

■■ 2. Step-by-Step Installation & Local Setup

Step 2.1: Prerequisites

Ensure **Node.js (v18.0.0 or higher)** and a local SQL database server (**XAMPP with MySQL** or **PostgreSQL**) are installed.

Step 2.2: Local Database Setup (Choose Option A or Option B)

Option A: Using XAMPP (MySQL / phpMyAdmin) - Recommended for Beginners

- Open the **XAMPP Control Panel**.
- Click **Start** next to **Apache** and **MySQL**.
- Open your web browser and navigate to: `http://localhost/phpmyadmin`
- In phpMyAdmin, click **Databases** from the top menu bar.
- In the **Database name** box, type `signlens_mobile`.
- Select `utf8mb4_general_ci` as collation and click **Create**.

Option B: Using PostgreSQL

- Open **pgAdmin** or your PostgreSQL command prompt (`psql`).
- Run the SQL command:

```
CREATE DATABASE signlens_mobile;
```

Step 2.3: Environment Configuration (.env File Setup)

Navigate to the `backend/` directory in Command Prompt / Terminal and create your local `.env` configuration file:

Windows (Command Prompt):

```
cd backend
copy .env.example .env
```

macOS / Linux / PowerShell:

```
cd backend
cp .env.example .env
```

Open backend/ .env in any text editor and configure your variables:

```
# Server Port
PORT=8001

# Secret Key for JWT Tokens
SECRET_KEY=super_secret_signing_key_for_signlens_mobile_app_1234567890

# Database Configuration (For PostgreSQL or MySQL)
PGHOST=localhost
PGPORT=5432
PGDATABASE=signlens_mobile
PGUSER=postgres
PGPASSWORD=postgres

# Machine Learning Subsystem Connection
MODEL_API_URL=http://127.0.0.1:8000
MODEL_API_WS_URL=ws://127.0.0.1:8000
MODEL_API_KEY=signlens_internal_api_key
```

Step 2.4: Install Dependencies & Run Database Migrations

In your terminal (inside the `backend/` folder), execute:

```
# 1. Install Node.js packages
npm install

# 2. Run Database Migrations (Creates tables: users, histories, lessons, quizzes, badges)
npm run migrate

# 3. Seed Database with initial sign language lessons and categories
npm run seed
```

■ 3. Starting the Backend Server

To start the backend in development mode (with auto-reload enabled):

```
npm run dev
```

Verification:

- Terminal will display: `Server running on port 8001.`
- Open your browser to: `http://localhost:8001/api-docs`
- You will see the **Swagger Interactive API Documentation** listing all available REST endpoints.

■ 4. Key Available API Routes

- **Auth Routes** (/api/v1/auth):
 - POST /register: User signup with bcrypt password hashing.
 - POST /login: User login returning stateless JWT token.
 - POST /google: Google OAuth ID Token verification.
 - **Prediction Proxy** (/api/v1/predict):
 - POST /predict?type=image: Forwards static image to Python ML server (Port 8000).
 - POST /predict?type=video: Forwards video clip to Python ML server.
 - WS /predict-stream: Duplex WebSocket streaming route.
 - **LMS Routes** (/api/v1/lessons, /api/v1/quizzes, /api/v1/badges):
 - GET /lesson-categories: Fetch sign language course categories.
 - GET /lessons/:id: Fetch sign language video tutorial details.
 - GET /quizzes/category/:categoryId: Interactive quiz questions.
 - **Translation History** (/api/v1/history):
 - GET /history: Fetch user's translation log history.
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■ 5. Common Backend Issues & Troubleshooting

■ Problem 1: connect ECONNREFUSED 127.0.0.1:5432 or MySQL Connection Error

- **Cause:** The database service (XAMPP MySQL or PostgreSQL) is not running.
- **Solution:** Open XAMPP Control Panel and click **Start** next to MySQL. Ensure the database signlens_mobile exists in phpMyAdmin.

■ Problem 2: Error: Migration directory is corrupt or missing

- **Cause:** knexfile.js cannot locate the migration folder.
- **Solution:** Ensure you execute npm run migrate from inside the backend/ directory root.

■ Problem 3: Error: ML backend returned an empty or invalid prediction response

- **Cause:** The Python FastAPI ML server (Port 8000) is not running.
- **Solution:** Start the ML server first by following the instructions in [model/README.md](file:///c:/Users/JOSHUA%20ASEMANI/Music/SignLens/SignLens/model/README.md).